Two Modified Low-light Image Enhancement methods

學生：方振輔 余宸逸 蔣繼霆
指導教授：柳金章 博士
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PairLIE – Introduction (1/3)

- Unsupervised method, trained by paired low-light images.
- By the Retinex theory, an image I can be decomposed into illumination L and reflectance R, i.e., $I = L \circ R$.



I_1 I_2

Fig. 1. A paired low-light images. I_1 and I_2 share the same reflectance with differ illuminations.

PairLIE – Introduction (2/3)

- A paired low-light images I_1 and I_2 share the same reflectance components with differ illuminations.
- The PairLIE [1] architecture contains three modules P-Net, L-Net, and R-Net.
- P-Net is designed to remove inappropriate features of the input image.
- L-Net and R-Net are used to evaluate the illumination and reflectance components of the input image, respectively.

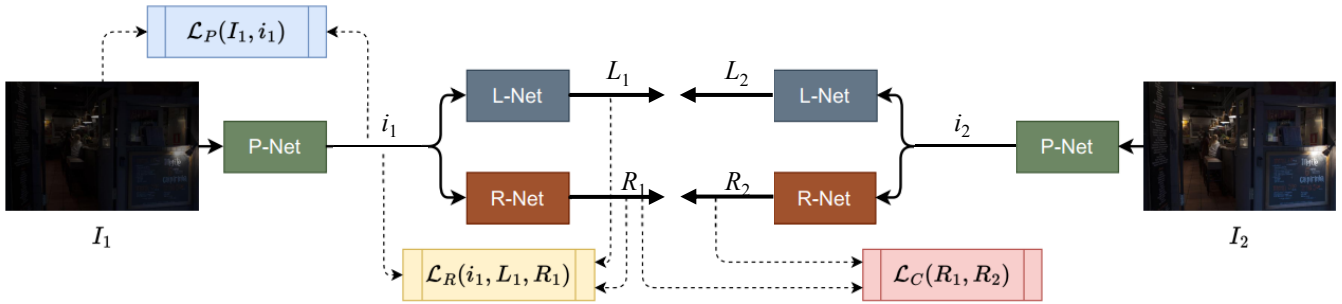


Fig. 2. PairLIE architecture in the training phase.

PairLIE – Introduction (3/3)

- In the testing phase, I is computed by $I = L \circ R$.
- Additionally, the illumination component L is adjusted via gamma correction before element-wise multiplication.

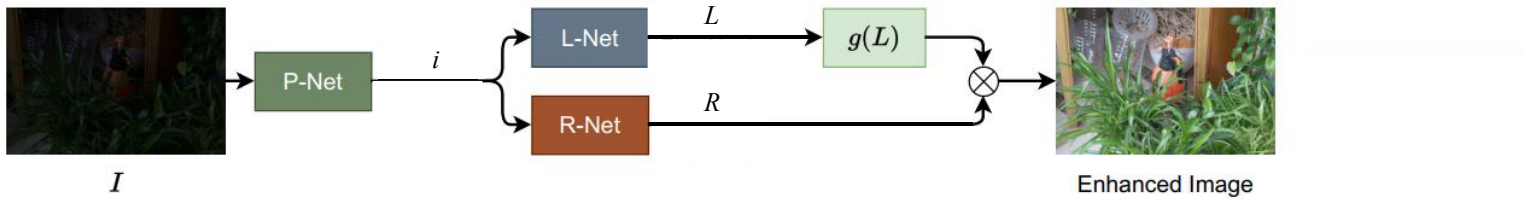


Fig. 3. The architecture of PairLIE in the testing phase.

g(.) : Gamma correction
ⓧ : Element-wise multiplication.

PairLIE – First Modified Method (1/3)

- In this study two modules, illumination estimator and illumination-guided attention block (IGAB), are added to assist R-Net to compute R.

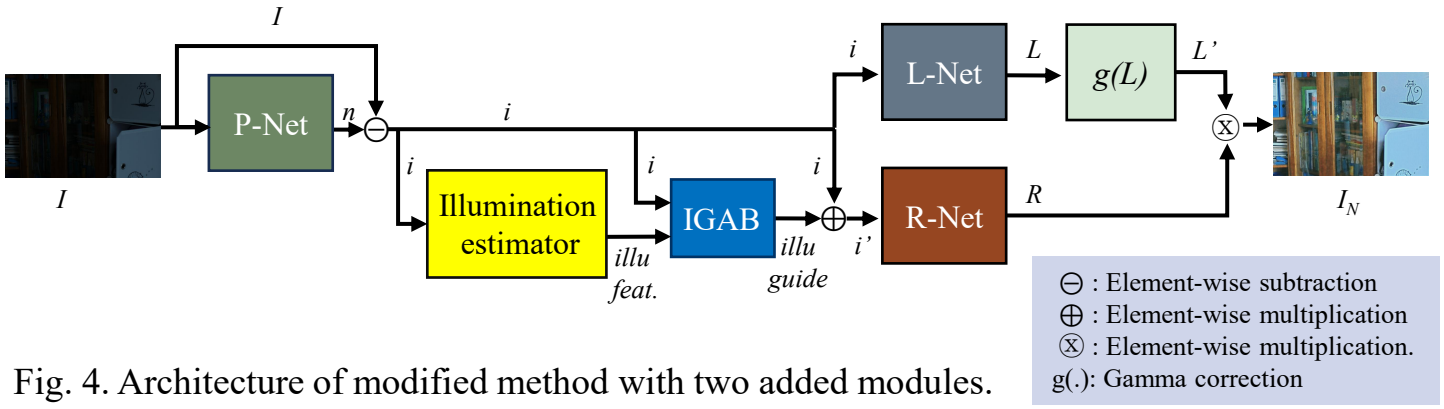


Fig. 4. Architecture of modified method with two added modules.

PairLIE – First Modified Method (2/3)

- In this study, illumination estimator analyzes the light distribution in the image and enhances the image illumination information.
- IGAB in Retinexformer [2] is a basic block of illumination-guided transformer (IGT) designed as a corruption restorer.

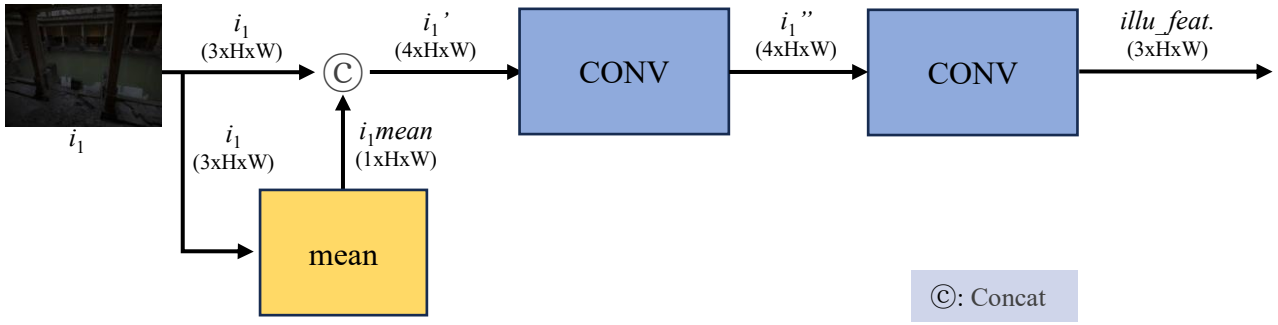


Fig. 5. Illumination estimator architecture.

PairLIE – First Modified Method (3/3)

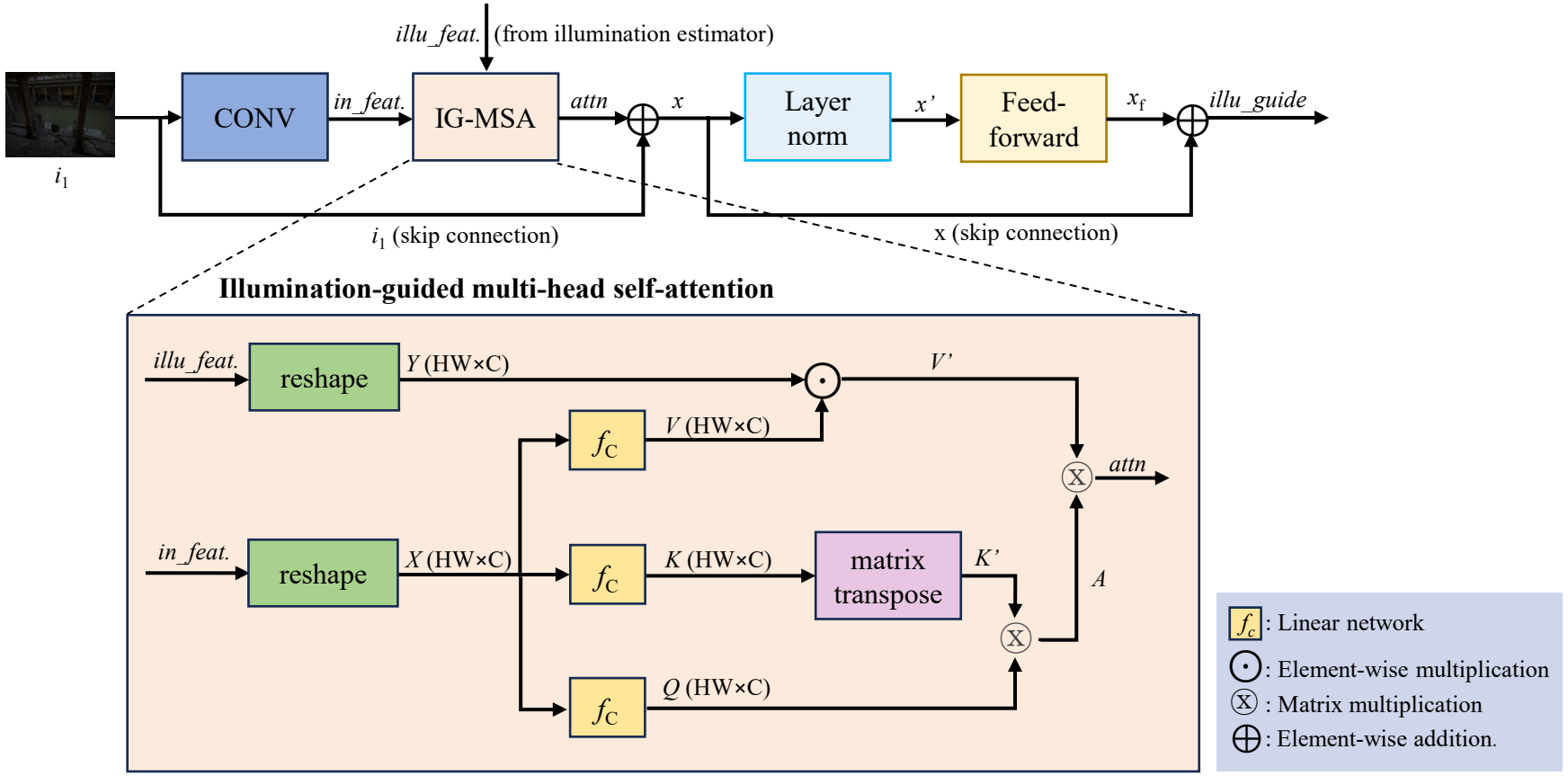


Fig. 6. The architecture of illumination-guided attention block (IGAB) and IG-MSA.

PairLIE – Experimental results (1/2)

Table 1. Performance comparisons between PairLIE and the first modified method.

Dataset	**LOL**		**SICE**	
Avg	PairLIE	Method 1	PairLIE	Method 1
PSNR (dB) (↑)	19.51	**19.66**	21.32	**21.54**
SSIM (↑)	0.736	**0.752**	0.840	**0.848**
LPIPS (↓)	0.248	**0.232**	0.216	**0.206**
Delta E (↓)	**10.80**	11.66	**7.835**	7.871

PairLIE – Experimental results (2/2)



Input PairLIE modified method 1 Ground Truth

Fig. 7. Visual comparisons between PairLIE and the first modified method.

RQ-LLIE – Introduction

- A supervised three-stage framework based on VQ-VAE [4].
- In stage 1, a codebook is constructed to learn normal-light image priors.
- In stage 2, learning from low-light images, a query module is designed to bridge the gap between low-light features and the normal-light codebook.
- In stage 3, a fusion branch is proposed to fuse low-light features and normal-light priors.
- Brightness-aware attention module improves the robustness of the brightness.

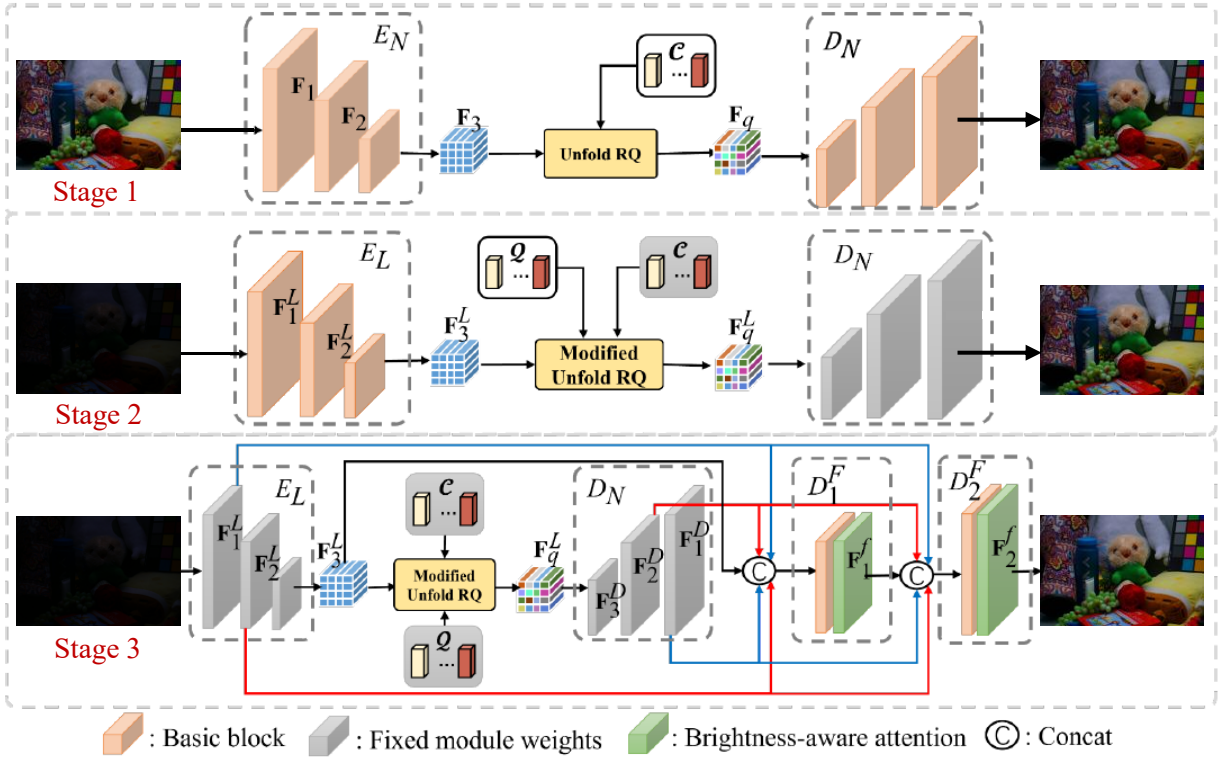


Fig. 8. Three-stage framework RQ-LLIE [3] for low-light image enhancement.

RQ-LLIE – Second Modified Method

- In this study, each input image will be converted to HSV color space and RQ-LLIE will followed by an HSV correction net.
- The hue component will not be changed, and the saturation and value components will be adjusted by S-Net and V-Net, respectively.
- Additionally, N-Net will convert the HSV image back to RGB color space.

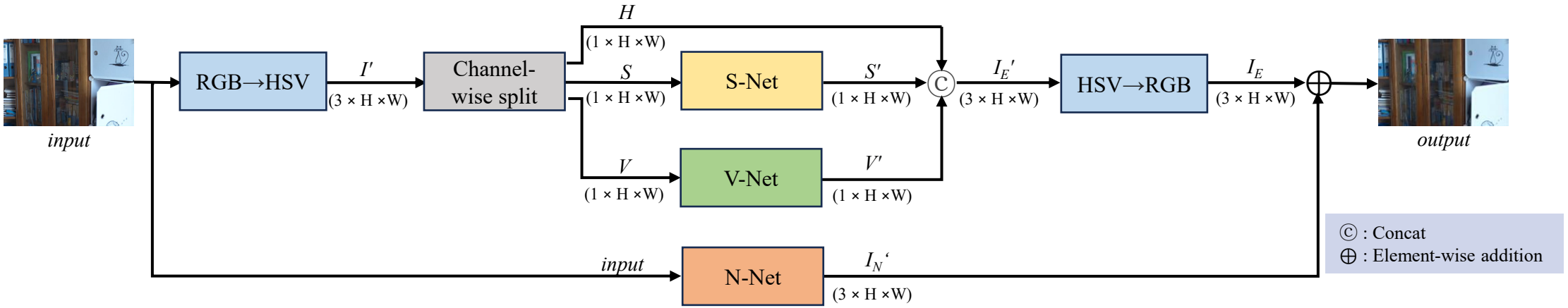


Fig. 9. The architecture of HSV correction net.

RQ-LLIE – Experimental results (1/2)

Table 2. Performance comparisons between RQ-LLIE and the second modified method.

Dataset	**LOLv1**	
Avg	RQ-LLIE	Method 2
PSNR (dB) (↑)	25.2407	**25.5454**
SSIM (↑)	0.8551	**0.8611**
LPIPS (↓)	0.1211	**0.1158**
Delta E (↓)	7.0609	**6.9087**

RQ-LLIE – Experimental results (2/2)



Input RQ-LLIE modified method 2 Ground Truth

Fig. 10. Visual comparisons between RQ-LLIE and the second modified method.

Concluding Remarks

- In this study, two modified low-light image enhancement methods are proposed.
- Based on the experimental results obtained in this study, the performances of the two modified methods are better than those of two comparison methods, PairLIE and RQ-LLIE, respectively.

References

[1] Z. Fu, Y. Yang, X. Tu, Y. Huang, X. Ding, and K. Ma, "Learning a simple low-light image enhancer from paired low-light instances," in *2023 IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, Vancouver, BC, Canada, 2023, pp. 22252-22261.

[2] Y. Cai, H. Bian, J. Lin, H. Wang, R. Timofte, and Y. Zhang, " Retinexformer: one-stage Retinex-based transformer for low-light image enhancement," in *2023 IEEE/CVF Int. Conf. on Computer Vision (ICCV)*, Paris, France, 2023, pp. 12470-12479.

[3] Y. Liu, T. Huang, W. Dong, F. Wu, X. Li, and G. Shi, "Low-light image enhancement with multi-stage residue quantization and brightness-aware attention," in *2023 IEEE/CVF Int. Conf. on Computer Vision (ICCV)*, Paris, France, 2023, pp. 12106-12115.

[4] A. Van Den Oord, et al, "Neural discrete representation learning," *Advances in Neural Information Processing Systems,* vol. 30, 2017.